

Initial Project Planning

Date: 15 July 2026

Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	User Authentication	USN-1	As a user, I can register an account using my email and password.	2	High	Team Member 1	16-07-2026	22-07-2026
Sprint-1	User Authentication	USN-2	As a user, I can securely log in to the HDI Prediction System.	2	High	Team Member 2	16-07-2026	22-07-2026
Sprint-1	Dataset Management	USN-3	As an administrator, I can upload and manage HDI datasets.	3	High	Team Member 3	16-07-2026	22-07-2026

Sprint-2	Data Preprocessing	USN-4	As a data analyst, I can clean and preprocess HDI data before model training.	4	High	Team Member 1	23-07-2026	30-07-2026
Sprint-2	Machine Learning Model	USN-5	As a researcher, I can train the machine learning model using HDI indicators.	5	High	Team Member 2	23-07-2026	30-07-2026
Sprint-3	HDI Prediction	USN-6	As a user, I can enter development indicators and receive an HDI prediction.	5	High	Team Member 3	31-07-2026	07-08-2026
Sprint-3	Dashboard	USN-7	As a user, I can view prediction results using interactive charts and graphs.	4	High	Team Member 4	31-07-2026	07-08-2026
Sprint-4	Country Comparison	USN-8	As a policymaker, I can compare HDI values of multiple countries.	4	Medium	Team Member 1	08-08-2026	15-08-2026

Sprint-4	Report Generation	USN-90	As a user, I can download prediction reports in PDF or Excel format.	3	Medium	Team Member 2	08-08-2026	15-08-2026
Sprint-5	Testing & Deployment	USN-10	As an administrator, I can deploy the application after testing and validation.	5	High	Entire Team	16-08-2026	25-08-2026

Sprint Summary

Sprint 1 – User Management & Dataset Setup

- User Registration
- User Login
- Dataset Upload
- Database Configuration

Sprint 2 – Data Processing & Model Training

- Data Cleaning
- Feature Engineering
- Model Training
- Model Evaluation

Sprint 3 – Prediction Module

- User Input Form
- HDI Prediction Engine
- Dashboard Development
- Data Visualization

Sprint 4 – Advanced Features

- Country Comparison
- Trend Analysis

- Report Generation
- Performance Optimization

Sprint 5 – Final Testing & Deployment

- System Testing
 - Bug Fixing
 - Documentation
 - Cloud Deployment
 - Final Demonstration
-

Project Timeline

Sprint	Duration	Deliverable
Sprint 1	1 Week	User Authentication & Dataset Setup
Sprint 2	1 Week	Data Preprocessing & ML Model
Sprint 3	1 Week	HDI Prediction System & Dashboard
Sprint 4	1 Week	Reports & Country Comparison
Sprint 5	1 Week	Testing, Documentation & Deployment

Project Goal

The project follows an **Agile development approach** with five sprints. Each sprint focuses on delivering a working feature, allowing continuous testing, feedback, and improvements. By the end of Sprint 5, the **Human Development Index (HDI) Prediction System** will provide secure user access, accurate machine learning-based predictions, interactive dashboards, downloadable reports, and a scalable deployment-ready solution.